

Connector OS

Complete Software

Architecture

Installable governance node: identity, memory, policy, audit, and effect mediation for agentic workloads — Seven Pillars solidification, Linux host enforcement, and court-grade claim discipline.

Includes

- Isometric internal system drawings
- Kernel · substrate · services census
- Effect exclusivity · ZT · microVM · LLM distrust
- Seven Pillars · eBPF · unbypassable profile
- Court/military claim grades · testing left

Truth rule

Coding of the governance bar is in-repo.
Military/court attach is evidence-gated.
Overclaim without signed verdict is refused.

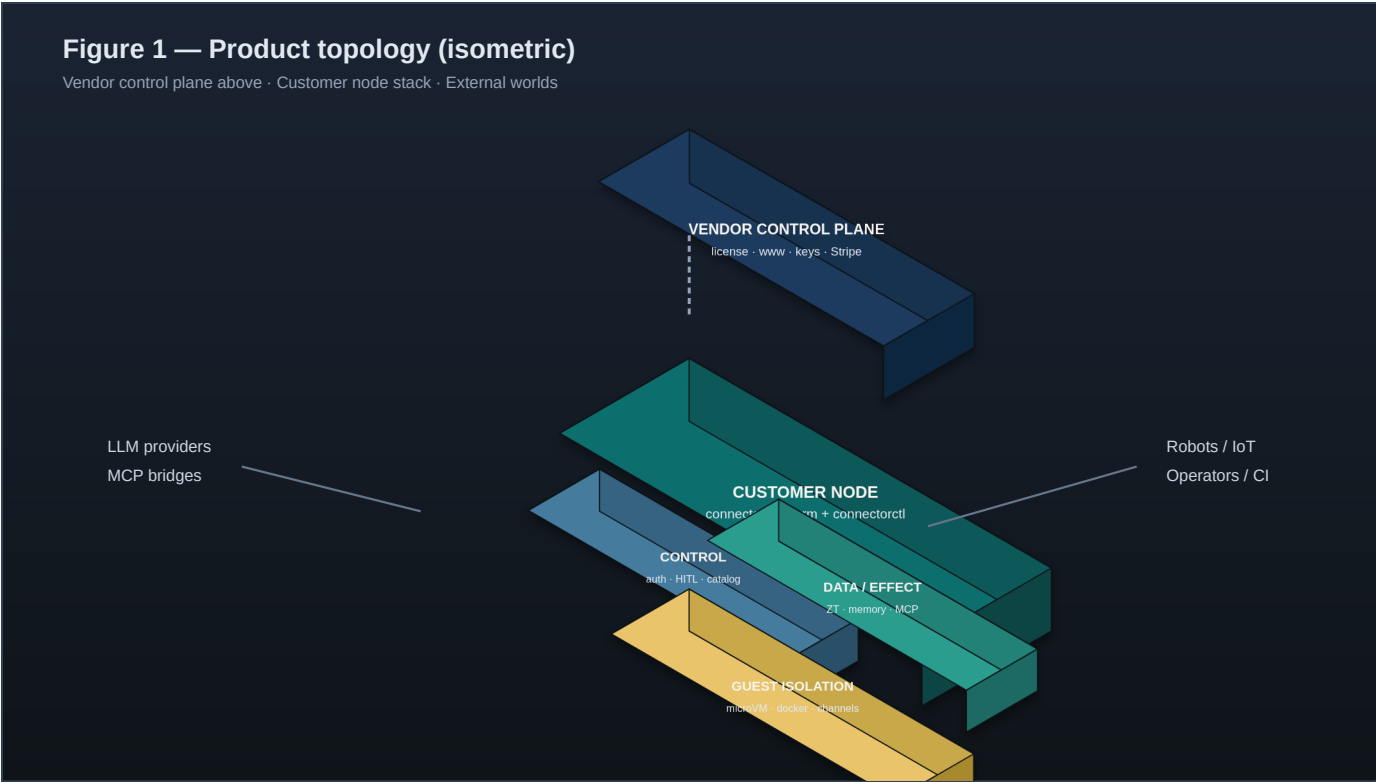
Slices: EFFECT_EXCLUSIVITY · ZT_HANDSHAKE · MICROVM_CHANNELS · PROBABILISTIC_LLM
Audience: architects · security reviewers · operators · node owners · counsel

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1. Product topology & mental model (with 3D drawing)
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10. Seven Pillars · unbypassable · court/military grades
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How to read: Figures are isometric projections of control, data/effect, and guest planes. Tables list every module an architect must locate. Closing pages state what we do *not* claim.

1. Product topology



Vendor plane is outside the customer install. One node = one governance kernel + optional host helpers.

What people install vs what we run

Role	Binary	UI	State
Customer node	connector-platform	Embedded Leptos dashboard	CONNECTOR_DATA_DIR
Customer CLI	connectorctl	—	systemd / HTTP
Vendor license	connector-license-server	www + admin	Postgres (typical)
Optional hub	connector-hub	—	.cpkg registry
Host egress helper	connector-kerneld	—	systemd/nft + eBPF mark-deny
Guest agent	connector-vm-agent	—	inside microVM

Mental model: workflows, .cpkg plugins, and world adapters are workloads on the substrate — not a kit of unrelated services. Vendor billing stays outside the on-prem governance kernel.

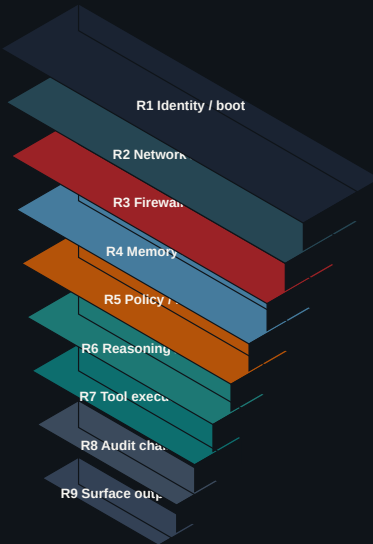
Three in-node planes

Plane	Secrets?	World sockets?	Primary code
Control	Yes (vault)	Yes (broker)	services/*, auth/*
Data / effect	vault: at last mile	Only broker + ZT	substrate/*, kernel/*, vac
Guest	No	No under distrust	plugin-runtime, microvm_tool_plane

2. Nine rings + intelligence overlay

Figure 2 — Nine rings + intelligence overlay (3D stack)

Concentric enforcement · each ring fail-closed · ACS / NSFS / grants sit on top



Intelligence overlay

- principal · ACS · NS FS
- world grants · AutonomyGateway
- agentic context · identity stack
- ZT handshake · DockLock
- effect exclusivity

LLM is probabilistic — not root

Ring	Concern	Primary code
1	Identity & boot	auth/, boot/, data/keys/
2	Network & gateway	router.rs, middleware
3	Firewall & guard	DevGuard, AutonomyGateway, graph firewall
4	Memory kernel	vac-core, services/memory*, Knot
5	Policy & governance	CCL/CLS, contracts, HITL, admission
6	Reasoning & LLM	gateway.rs, agentic inject
7	Tool execution	ZT → microVM / MCP / CONP
8	Audit chain	HMAC journal, receipts, forensics
9	Surface output	SOE / role redaction

Intelligence overlay (on top of rings)

Piece	Module	Binds
Principal	kernel/agent_principal.rs	Cryptographic agent identity
Envelope	agent_identity_envelope.rs	Setup/activate, grants
ACS	acs.rs	Character / cage posture
NS FS	nsfs.rs	/m /k /p /share per pid
World grants	world_gateway.rs	(agent × address)

AutonomyGateway	action_binding.rs	Allow Ask Block
Agentic context	substrate/agentic_context.rs	Who-am-I into talk

3. Effect path & exclusivity



Effect exclusivity standard

Under hardened posture an agent **cannot** produce a consequential effect except through the Connector effect mediator.
Production: `CONNECTOR_EFFECT_EXCLUSIVITY=1`.

Alternate path	How it is closed
Raw network / TCP / HTTP	L7 + MCP egress + Ring-1; guest deny_all / vsock-only
Ungoverned shell / subprocess	Isolation ≠ subprocess; shell I/O via microVM tool plane
In-process ToolDispatch / MCP	Isolation bar + ZT; no ALLOW_IN_PROCESS_EFFECTS
Secrets in guest env	Cage strip + isolation membrane
Ungoverned memory	governed_effect + identity stack + Ring-1
A2A without grant	Inter-intelligence or world address grant

APIs: `GET .../effect-exclusivity/status` · `POST .../probe` · `adversarial script platform/scripts/effect-exclusivity-adversarial.sh`

Break-glass (weakens claim): `CONNECTOR_ALLOW_IN_PROCESS_EFFECTS=1`, `CONNECTOR_ALLOW_GUEST_EGRESS=1`

4. Zero-trust handshake & probabilistic LLM

ZT handshake (blockchain-grade tool binding)

Policy admission is not enough if the model can call MCP/HTTP directly. Connector is the **only signer** of tool authority. Session key is **never** given to the LLM.

Property	How
Genesis immutability	SHA-256 genesis + node Ed25519
Session secrecy	Rederived from node signature of zt-session handshake_id
Chain continuity	Each ticket prev_hash = previous block_hash
Single-use	Nonce spent; replay denied
Forgery resistance	HMAC-SHA256 MAC + node signature

Env: CONNECTOR_ZT_HANDSHAKE=1 · Module: kernel/zt_handshake.rs · admit_tool_effect on every ToolDispatch

Probabilistic LLM guard

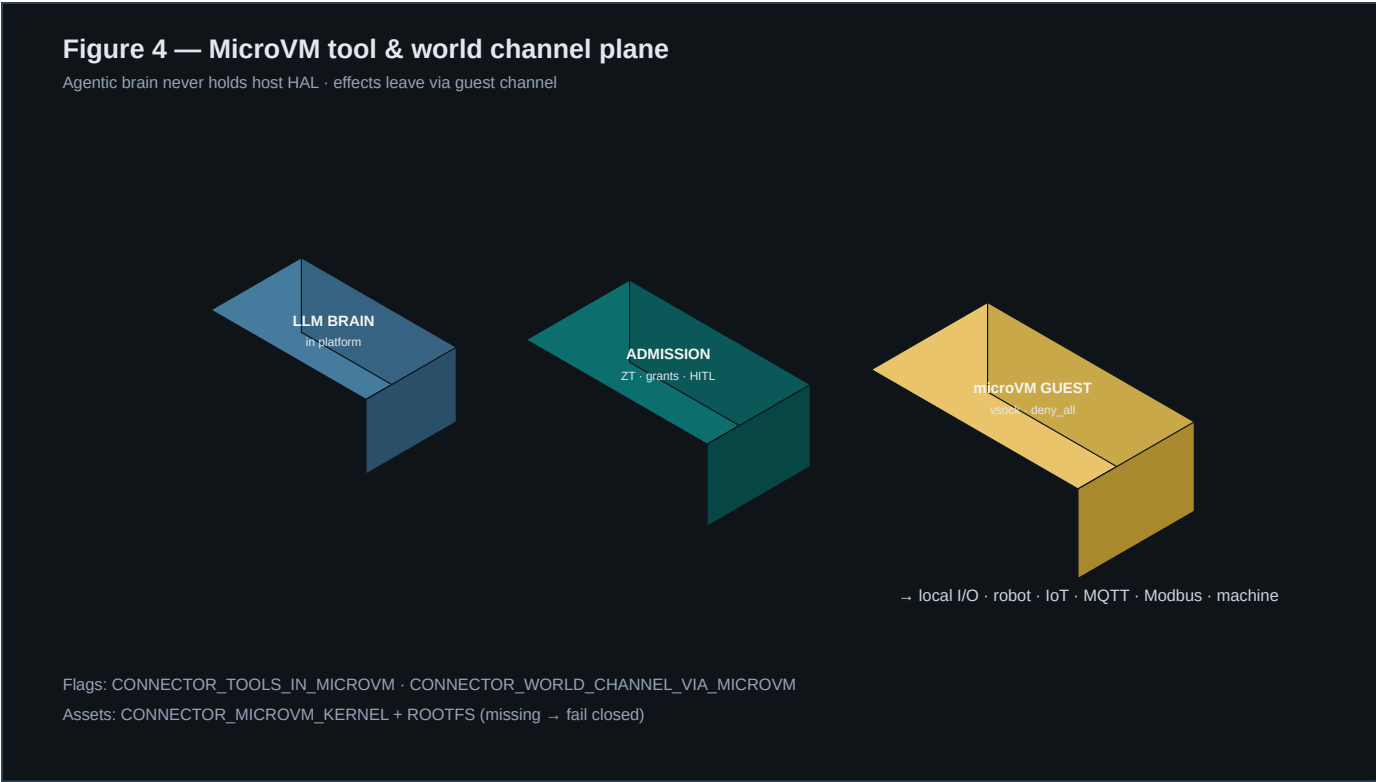
Connector does not treat the model as ready for unaugmented autonomy. Identity, memory, character, knowledge, HITL, and parameters stay **Connector-owned**.

Event	Action
Bypass of Connector rule (forged ticket, in-process path, exclusivity skip)	Quarantine → human retrieval (unquarantine HITL)
Failure of identity / memory / character / knowledge / HITL / parameters	Mandatory human approval

Env: CONNECTOR_LLM_DISTRUST=1 · Module: substrate/probabilistic_llm.rs · Status: GET .../probabilistic-llm/status

Pair the runtime membrane with docker_lab / microVM so guests physically cannot open sockets around Connector.

5. MicroVM tool & world channels



Channel	Examples	Env
Local I/O	shell, read/write, exec	CONNECTOR_TOOLS_IN_MICROVM=1
Robotics	robot:..., HAL, ROS	CONNECTOR_WORLD_CHANNEL_VIA_MICROVM=1
IoT	iot:..., sensors/actuators	same
MQTT / Modbus	mqtt://..., modbus:...	same
Machine / device	machine:..., CNC, PLC	same

Remote MCP HTTPS may remain a **host broker** when `CONNECTOR_ALLOW_HOST_MCP_BROKER=1` — it still cannot do local I/O or physical channels on the host.

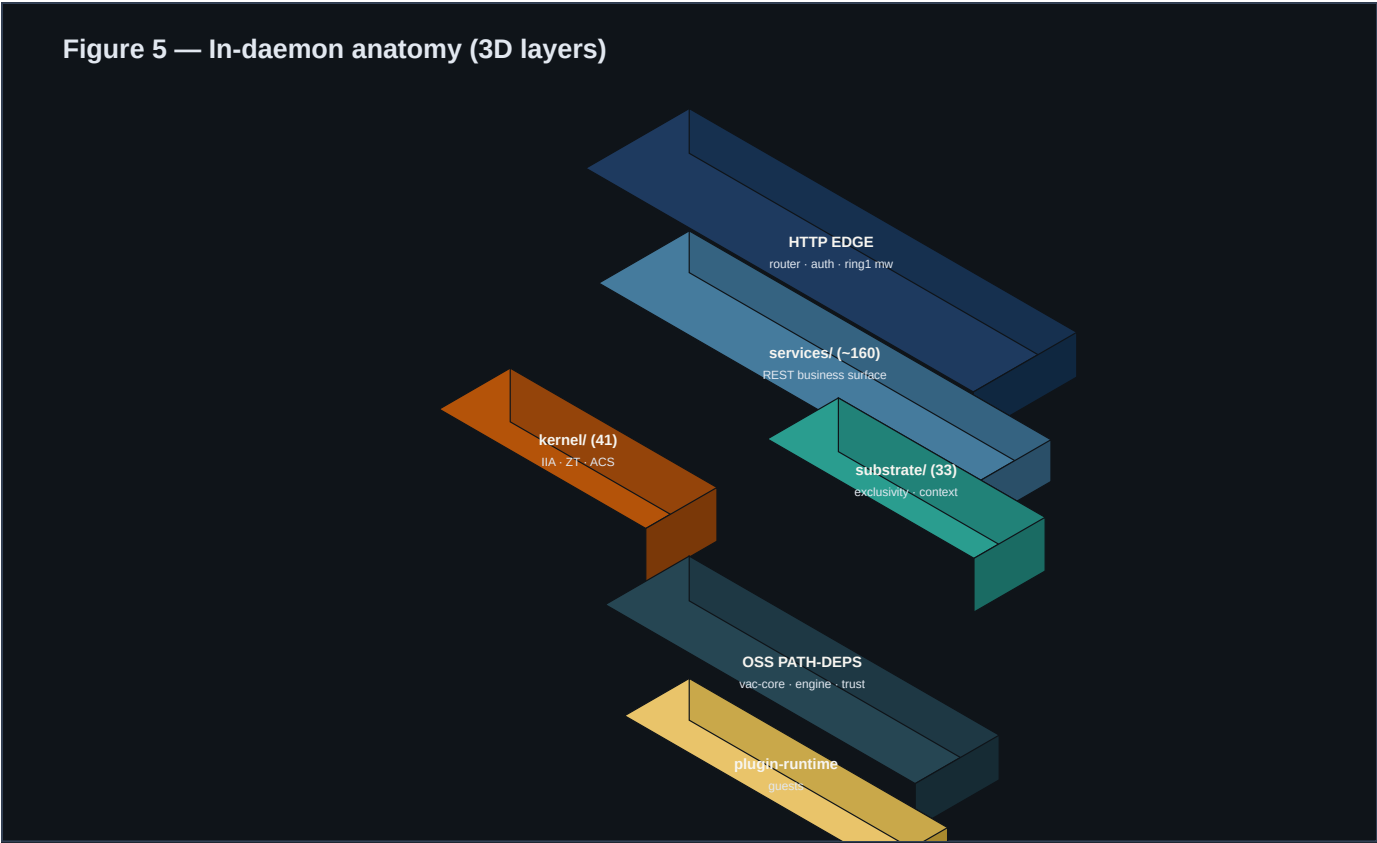
APIs: GET/POST `.../microvm-tools/*` · CONP command routes physical entities via microVM when flags on · Assets: KERNEL + ROOTFS required

Isolation backends

Backend	When	Under distrust
subprocess	Lab/dev	Weak — host process
docker_lab	Lab / optional	--network none
microvm	Production hardening	vsock-only, no TAP
wasm	Constrained	WASI, no TCP
internal Connector OS Architecture	Break-glass / lab	In-process

6. In-daemon anatomy & kernel

Figure 5 — In-daemon anatomy (3D layers)



HTTP edge → services (~160) → kernel (41) + substrate (33) → OSS vac/engine → plugin-runtime guests

Kernel encyclopedia (selected critical modules)

Module	Responsibility
operating_layer / aios	AIOS sockets; Memory OS syscalls
acs / nsfs	Character surface; per-agent FS homes
address_cage / world_gateway	Typed world addresses; owner grants
admission_layers / action_binding	Root/Cone/App; AutonomyGateway
agent_principal / envelope	Mint principal; setup/activate
zt_handshake	Ed25519 + hash-chained tool tickets
docklock / ring1_context	ExecutionQuantum; per-request quantum
credential_proxy / mission_journal	vault: resolve; durable steps
membrane_posture / isolation_tiers	Refuse if backends missing; tier ladder
forensics / decision_trace	Receipt chains; court packages
share_portal / council	Cross-agent share; sibling intelligences

Full 41-module table: CONNECTOR_FULL_ARCHITECTURE.md Part V

Substrate (critical)

Module	Responsibility
governed_effect	Unified effect admission
effect_exclusivity	Close six alternate paths
probabilistic_llm	Quarantine / mandatory HITL
agentic_context / identity_stack	Who-am-I inject; pillar gate
microvm_tool_plane	I/O + physical channel routing
cage_security / egress_policy / cfni	Cage caps; L7/MCP; hop headers
agent_lifecycle_gate / progeny	Only path to lifecycle syscalls

7. Services, binaries, OSS, plugins

Services groups (~160 modules)

Group	Examples
Agents / IIA	agents, agent_identity, admission, iia_runtime, world_gateway, council, missions
Memory / knowledge	memory, memory2, memory_graph, knowledge_*, assets, cognitive
Tools / gateway	tools, gateway, anthropic_gateway, protocols, conp_protocol, mcp_hosting
Plugins / workflows	plugin_*, workflow_*, cls, apps_catalog, cage_proof
DevGuard / TT / WC	devguard*, fs_guard, exec_guard, tracetramp_proxy, witnessctl_proxy
Runtime / mesh	runtime_*, mesh_*, ha_federation, kernel_host
Audit / proof	actionlog, proof*, forensics, compliance, books, verify
Business	licensing, billing, playground*, settings_*, setup

Platform crates (census)

Package	Binary	Job
connector-platform	platform + connectorctl	Commercial kernel + CLI
connector-plugin-runtime	lib	Isolation backends + membrane
connector-microvm / vm-agent	vm-agent	Firecracker types + guest PID1
connector-kernelld	kernelld	systemd/nft reconcile + eBPF cgroup/skb load
connector-cpkg / hub	hub	.cpkg format + registry
connector-license-server	license-server	Vendor keys / portal
connector-ui / www / admin / trial	Uis	Dashboard & vendor surfaces

OSS path-deps & first-party plugins

OSS: vac-core/store/crypto/... · connector-engine/api/protocols/trust/caps · aapi-*

Plugins: TraceTramp · WitnessCtl · DevGuard · Conductor · AgentLoop · AgentPassport · Engram · LedgerLens · Relay

8. Data plane, presets, hardening

Persistence (CONNECTOR_DATA_DIR, default ./data)

Path	Role
engine.db	SQLite — engine/audit/secrets
kernel.redb	redb — memory kernel packets, agents
users.db	Auth/users
keys/	Signing / verifying keys
workflows/, catalog/	Workflow catalog
plugins/cpkg_store	Installed .cpkg rollouts
nsfs/{pid}/	Per-intelligence filesystem homes
object_fabric_cas/	CAS blobs

Presets (CONNECTOR_PRESET)

Class	Values	Posture
Lab	local, ci, docker-local, plugin-workbench	Stub LLM OK, weak isolation
Prod-like	staging, production, airgap, defense-strict, unbyypassable, edge-satellite	Hardening + microVM + sandbox bar
Trial	playground, trial, saas-trial	TTL, caps — not enterprise secret gates

Production hardening defaults (non-exhaustive)

- Ring-1 / QPR / DockLock / HITL · setup gate · ban anon gateway
- Kernel + Landlock fail-closed · L7 + MCP egress · CFNI · flow lease enforce
- Effect exclusivity · ZT handshake · LLM distrust · agentic context require
- Isolation microVM · tools/world via microVM · guest deny_all · vsock tickets
- CONNECTOR_SANDBOX_UNBYPASSABLE · matrix host egress · eBPF require
- Profile file: platform/deploy/unbypassable.env

Core ports

Env	Default
CONNECTOR_PORT	9091
CONNECTOR_PROTOCOL_PORT	9092 (MCP/A2A/...)
CONNECTOR_UI_RPC_PORT	9093

9. API surface & hard problems

HTTP mounts

Mount	Role
/api/v1/*	Primary REST (auth-gated)
/api/v2/*	Simplified V2
/v1/chat/completions, /v1/messages	OpenAI / Anthropic gateway
/plugin/<slug>/*	Cage reverse-proxy
Protocol port	MCP / A2A / ACP / ANP / AP2
UI-RPC port	Dashboard WebSocket JSON-RPC

Hard problems → solvers

Hard problem	Pieces
LLM invents who it is	principal, ACS, agentic_context, identity_stack
Model calls tools directly	ZT + exclusivity + guest deny_all
Identity collapse across agents	NS FS + share portal + A2A grants
Host ambient authority	address_cage + world grants + eBPF/nft cut
Robot/IoT on control plane	microvm_tool_plane + vsock tickets
Sandbox bypass / revoke race	sandbox_unbypassable + atomic_revoke
Probabilistic bypass / drift	quarantine + mandatory HITL
Resume re-fires effects	mission_journal + digests
Secrets in the cage	credential_proxy + membrane

Talk path (summary)

Auth → budgets / injection → inject who-am-I + agentic context + memory → incomplete stack under REQUIRE → HITL → LlmRouter → provider → audit + recall.

10. Seven Pillars · unbypassable · court grades

Status discipline: SHIPPED_VERIFIED | IMPLEMENTED_GATED | PLANNED | ASPIRATIONAL. Living table: SEVEN_PILLARS_STATUS.md. PDF checklist coding is complete; attach proofs remain host-gated.

Seven Engineering Solidification Pillars

#	Pillar	Coding bar
1	Agent identity & cognitive confinement	Principal · ACS · agentic context · soak non-collision
2	OS-grade L1/L2/L3 + kerneld	Landlock FC · cgroup bind · eBPF mark-deny · desired-vs-applied
3	Cryptographic HITL / exact-action	EffectAuthorization · one-byte deny · nonce/expiry
4	Network identity & flow admission	Flow lease · socket binding · peer overlay · raw-socket deny
5	Invariant effect routing	EffectEnvelope · exclusivity · governed_effect
6	Isolation contract	Manifest · vsock HMAC tickets · measured microVM · atomic revoke
7	cpkg / CLS / AAPI / CNP	Security contract · compile · wire signing

Court / military claim grades

Grade	Meaning	Typical verdict
GOVERNANCE	Authority model + crypto HITL/effect evidence	Met when unit/adversarial green
HOST_LAB	Fail-closed host lab (nft/bpftool; eBPF load <i>or</i> REQUIRE fail-closed)	Met on locked profile lab
MILITARY_COURT	eBPF <i>loaded</i> · Landlock visible · measured microVM · kerneld Active · no break-glass	Refused until attach evidence

Registry: security_claim_registry.json · Binder: court-grade-evidence-bind.sh · Overclaim: CLAIM_MILITARY_COURT=1 exits non-zero when unmet.

Linux host plane (coded)

Piece	Path
eBPF mark-deny	platform/ebpf/connector_mark_deny.bpf.c
kerneld load/status	connector-kerneld_ebpf-load status deny-mark
Unbypassable gate	substrate/sandbox_unbypassable.rs
Locked profile	platform/deploy/unbypassable.env
Host lab	seven-pillars-host-adversarial-lab.sh

11. Testing remaining & human trust

Honest split: governance and fail-closed *paths* are coded. What remains is host/adversarial *evidence* so attach claims can become SHIPPED_VERIFIED / MILITARY_COURT.

Host evidence backlog (testing left)

ID	Proof needed	Gate
H1	eBPF program loaded + pinned (CAP_BPF)	Military/court
H2	Landlock ABI visible + applied in opener process	Military/court
H3	Measured microVM kernel/rootfs at every tool spawn	Military/court
H4	kerneld HostApplyState::Active (not Simulated)	Military/court
H5	Packet lab: marked cgroup egress drop	Release
H6	Retain CONNECTOR_AUDIT_HMAC_KEY + court verdict package	Custody

Product / partner backlog

ID	Status	Bar
T2	Complete	eBPF connect4 + nft NAT + connector-egress-proxy TLS terminate (Connector CA)
T4	Solidified · RG-09	Idempotent journal + refuse Pending re-fire + abandon on resume
T6	Complete	ROS/Modbus/MQTT/TCP wire HAL + SIL safety interlock (partner certifies SIL body)
T7	Solidified	T00LS_IN_MICROVM_STRICT · host MCP broker off in unbypassable
T8	Solidified	/apps · /apps/parity · workflow catalog routes

Host attach: seven-pillars-host-attach-proofs.sh → military_court_ready=true when eBPF loaded + Landlock applied + measured microVM + TLS/SIL coded.

Real limitations (do not overclaim)

- Membrane ≠ full OS syscall proof until H2/H5 pass on the target host
- LLM forward-pass still runs in connector-platform (distrusted)
- Remote MCP ignoring ZT needs network isolation
- MicroVM needs operator-supplied measured kernel/rootfs
- eBPF is a real loader — attach still requires CAP_BPF / bpffs
- CONP lab HAL is echo (SIL is partner) · CNP ENABLE is not a live event bus
- Break-glass flags weaken the claim · court binder refuses military overclaim

Humans always trusted — by design

Role	Owns
Identity author	Purpose, character, denied ops
Grant author	Which addresses an intelligence may touch

Approver	Ask digests + unquarantine after bypass
Trust-anchor operator	Node keys, license roots, guest images, audit HMAC
Physical safety owner	Robots / CNC / IoT beyond lab echo

The LLM is never: root of identity · signer of tool authority · issuer of world grants · judge of quarantine · source of secrets.

Closing claim

Connector OS is an installable governance node. Under production / defense-strict / unbypassable hardening it closes alternate effect paths, distrusts the probabilistic model, projects policy to Linux (systemd/nft/eBPF connect4 + mark-deny), redirects egress into a Connector proxy hop, and pushes tool/physical I/O into microVM channels with vsock tickets and partner HAL wire adapters. It claims **GOVERNANCE** and **HOST_LAB** when evidence is green. It does **not** claim **MILITARY_COURT** without signed attach evidence, SIL robotics certification, live CNP bus, or model autonomy.

Appendix A — Full kernel module list

Module	Job
operating_layer	AIOS three sockets; vendor-blind absorb
aios	Memory OS syscalls core/recall/archival
intelligence_matrix	CDMI Albus SP·WM·VJ·BG
acs / nsfs	Character; per-pid FS
address_cage / address_contracts / address_dac_api	Typed addresses; DAC; HTTP
world_gateway / admission_layers	Grants; Root/Cone/App
action_binding	AutonomyGateway + digests
agent_principal / foundation / envelope	Identity mint & activate
compliance_contract / continuity	Auditor bind; reality manifest
intelligence_spec / purge	Declare-apply; kill cleanup
share_portal / council	Cross-agent; siblings
docklock / ring1_context	Quantum + cage env
zt_handshake / credential_proxy	Tickets; vault materialize
mission_journal / membrane_posture / isolation_tiers	Resume safety; readiness; tiers
matrix_isolation / matrix_host_egress / agent_cgroup	Continuity break; nft/eBPF cut; cgroup bind
isolation_manifest / effect_authz	Desired/measured isolation; EffectAuthorization
vault_seal / fabric_task	AES vault; A2A task SoT
decision_trace / forensics* / forensic_*	Traces; receipts; packages; rollups
agent_chat / agent_explain / iia_llm_inject	Talk persist; explain; inject
node_fabric / witnessctl_align	Topology honesty; WC align

Substrate full list

governed_effect · admission_gate · admission_matrix · admission_status · identity_stack · agentic_context · probabilistic_llm · effect_exclusivity · microvm_tool_plane · sandbox_unbypassable · atomic_revoke · effect_envelope_bus · seven_pillars_soak · cage_security · egress_policy · cfni · flow_lease · graph_firewall · sgke_gate · agent_lifecycle_gate · agent_progeny · outbound · proxy_auth · artifact_log · causal · usage_event · usage_receipt · cnp_edge · durability · memwrite_durability · knot_rebuild · handoff_queue · projection · retention · status · docklock_bridge

Appendix B — Repo anatomy & talk/plugin paths

Top-level repository

Path	Role
platform/	Commercial OS (ships)
platform/ebpf/	Host eBPF programs (mark-deny)
platform/deploy/unbypassable.env	Locked defense profile
oss/	Path-deps: connector, vac, aapi
plugins/	First-party AGOS products
agos-abi/ · agos-sdk/ · cargo-connector/	Plugin authoring
docs/ · lab/ · advanced-lab/	Library; non-prod labs
data/ · dist/ · vendor/	State; package; Firecracker assets

Plugin / workflow path

.cpkg / workflow.yaml → Ed25519 verify → IsolationRuntime → plugin-runtime spawn **or** CLS lease runner → catalog row (id · status · PID · URI).

Supporting in-daemon packages

auth · api_v2 · operator · boot · cls · cnp · intelligence_admission · quanta_polar · protocol_gateway · internal_dns · ui_rpc · middleware · kms · knowledge · knot · distributed · export · compliance · data · proof · policy · privacy

System of record: When you add a kernel/substrate/service module, crate, binary, or preset, update platform/docs/arch/CONNECTOR_FULL_ARCHITECTURE.md and SEVEN_PILLARS_STATUS.md in the same PR, then regenerate this PDF via bash platform/docs/arch/pdf-assets/rebuild_pdf.sh.

End of booklet · Connector OS Full Software Architecture · Encyclopedia edition · Seven Pillars + court-grade honesty